

Quiz 2

Tick the statements that are true and leave unticked those that are false.

Each statement is worth 1 point for a correct answer.

Statements with a * are bonus.

Your name: _____

Which of the following statements are correct?

- ☒ Every finite language is decidable.
- ☐ The property ‘ π has a **while** loop’ is semantic.
- ☒ If a language L is decidable, then the language $\Sigma^* \setminus L = \{w \in \Sigma^* \mid w \notin L\}$ is also decidable.
- ☐ There is a binary encoding of reals, such that $\{\vec{v} \in \mathbb{R}^2 \mid v_1 - v_2 > 0\}$ is enumerable.
- ☐ Rice’s theorem says that computers can’t cook rice al dente.
- ☒ A language can be enumerable but not decidable.
- ☐ Every property P of programs, such that $P \neq \emptyset$ and $P \neq \Sigma^*$, is semantic.
- ☒ There is a binary encoding of rationals, such that $\{\vec{v} \in \mathbb{Q}^3 \mid v_1 + v_2 = v_3\}$ is enumerable.
- ☒ Every language is countable (as a set of words).
- ☒ The property ‘ π sorts the input array in the ascending order’ is undecidable.
- ☐ Every infinite language is undecidable.
- ☐ The property ‘on all inputs, π outputs «Hello, World!»’ is decidable.
- ☒ The language $\{w \in \{0, 1\}^* \mid w \text{ contains exactly three symbols '1'}\}$ is decidable.
- ☐ Every property of programs is semantic.
- ☐ The property ‘ π has a **for** loop’ is semantic.
- ☒ *If both L and $\Sigma^* \setminus L$ are enumerable then L is decidable.
- ☒ *The universal property $P = \Sigma^*$ (every program has it) is semantic.

How hard is this quiz, from 0 (very easy) to 10 (very hard)?

Explanations

- ⊠ *Every finite language is decidable.* Compare the input with each of the finitely many words in the language. This procedure always halts, so it decides the language.
- *The property ‘ π has a `while` loop’ is semantic.* This is a syntactic property: two programs can compute the same function even if only one contains such a loop.
- ⊠ *If a language L is decidable, then the language $\Sigma^* \setminus L = \{w \in \Sigma^* \mid w \notin L\}$ is also decidable.* Exchange the yes and no answers of a program that decides L .
- *There is a binary encoding of reals such that $\{\vec{v} \in \mathbb{R}^2 \mid v_1 - v_2 > 0\}$ is enumerable.* An encoding by finite binary words can encode only countably many objects, whereas $\mathbb{R}$ is uncountable.
- *Rice’s theorem says that computers cannot cook rice *al dente*.* Rice’s theorem says that every non-trivial semantic property of programs is undecidable.
- ⊠ *A language can be enumerable but not decidable.* The language HALT is enumerable by dovetailing but undecidable.
- *Every property P of programs such that $P \neq \emptyset$ and $P \neq \Sigma^*$ is semantic.* Non-triviality does not imply that a property is semantic; having a `while` loop is a counterexample.
- ⊠ *There is a binary encoding of rationals such that $\{\vec{v} \in \mathbb{Q}^3 \mid v_1 + v_2 = v_3\}$ is enumerable.* Rational numbers admit an effective binary encoding. The relation is decidable by checking the encoded equality, and every decidable language is enumerable.
- ⊠ *Every language is countable (as a set of words).* The set Σ^* is countable, and every language is a subset of Σ^* .
- ⊠ *The property ‘ π sorts the input array in the ascending order’ is undecidable.* The property is semantic and non-trivial, so it is undecidable by Rice’s theorem.
- *Every infinite language is undecidable.* For example, Σ^* is infinite and is decided by a program that always accepts.
- *The property ‘on all inputs, π outputs «Hello, World!»’ is decidable.* This is a non-trivial semantic property, so it is undecidable by Rice’s theorem.
- ⊠ *The language $\{w \in \{0,1\}^* \mid w \text{ contains exactly three symbols ‘1’}\}$ is decidable.* Scan the word while counting its 1s, and accept exactly when the final count is three.
- *Every property of programs is semantic.* Some properties, such as the presence of a particular programming construct, are syntactic.
- *The property ‘ π has a `for` loop’ is semantic.* This is syntactic: programs computing the same function need not use the same programming constructs.
- ⊠ **If both L and $\Sigma^* \setminus L$ are enumerable, then L is decidable.* This is Post’s theorem.
- ⊠ **The universal property $P = \Sigma^*$ is semantic.* Programs computing the same function certainly agree on a property that every program has. The property is semantic but trivial.